2. a. Dik: $a = 2$, $b = 9$

Matriks kunci A

$$A = \begin{bmatrix} a+1 & b & 2 \\ 1 & a+2 & b \\ b & 1 & a+3 \end{bmatrix}$$

$$A = \begin{bmatrix} 2+1 & 9 & 2 \\ 1 & 2+2 & 9 \\ 9 & 1 & 2+3 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 9 & 2 \\ 1 & 4 & 9 \\ 9 & 1 & 5 \end{bmatrix}$$

SiDU

Dit : hitung det(A) menggunakan ekspansi baris pertama

Dij : $A = \begin{bmatrix} 3 & 9 & 2 \\ 1 & 4 & 9 \\ 9 & 1 & 5 \end{bmatrix}$

- M_{11} (hapus baris 1 kolom 1)
  $\begin{bmatrix} 4 & 9 \\ 1 & 5 \end{bmatrix} = 4\times5 - 9\times1 = 20-9 = 11$
- M_{12} (hapus baris 1 kolom 2)
  $\begin{bmatrix} 1 & 9 \\ 9 & 5 \end{bmatrix} = 1\times5 - 9\times9 = 5-81 = -76$
- M_{13} (hapus baris 1 kolom 3)
  $\begin{bmatrix} 1 & 4 \\ 9 & 1 \end{bmatrix} = 1\times1 - 4\times9 = 1-36 = -35$
- M_{21} (hapus baris 2 kolom 1)
  $\begin{bmatrix} 9 & 2 \\ 1 & 5 \end{bmatrix} = 9\times5 - 2\times1 = 45-2 = 43$
- M_{22} (hapus baris 2 kolom 2)
  $\begin{bmatrix} 3 & 2 \\ 9 & 5 \end{bmatrix} = 3\times5 - 2\times9 = 15-18 = -3$
- M_{23} (hapus baris 2 kolom 3)
  $\begin{bmatrix} 3 & 9 \\ 9 & 1 \end{bmatrix} = 3\times1 - 9\times9 = 3-81 = -78$
- M_{31} (hapus baris 3 kolom 1)
  $\begin{bmatrix} 9 & 2 \\ 4 & 9 \end{bmatrix} = 9\times9 - 2\times4 = 81-8 = 73$
- M_{32} (hapus baris 3 kolom 2)
  $\begin{bmatrix} 3 & 2 \\ 1 & 9 \end{bmatrix} = 3\times9 - 2\times1 = 27-2 = 25$
- M_{33} (hapus baris 3 kolom 3)
  $\begin{bmatrix} 3 & 9 \\ 1 & 4 \end{bmatrix} = 3\times4 - 9\times1 = 12-9 = 3$

Gunakan tanda : $(+ - +)$

$C_{11} = +M_{11} = +11$

$C_{12} = -M_{12} = -(-76) = 76$

$C_{13} = +M_{13} = -35$

$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$

$\det(A) = 3(11) + 9(76) + 2(-35)$

$\det(A) = 33 + 684 - 70 = 647$

b. Tabel matriks minor M_{ij}

M_{ij}	Perhitungan Minor	Hasil
M_{11}	$\begin{bmatrix} 4 & 9 \\ 1 & 5 \end{bmatrix} = (4\times5) - (9\times1)$	11
M_{12}	$\begin{bmatrix} 1 & 9 \\ 9 & 5 \end{bmatrix} = (1\times5) - (9\times9)$	-76
M_{13}	$\begin{bmatrix} 1 & 4 \\ 9 & 1 \end{bmatrix} = (1\times1) - (4\times9)$	-35
M_{21}	$\begin{bmatrix} 9 & 2 \\ 1 & 5 \end{bmatrix} = (9\times5) - (2\times1)$	43
M_{22}	$\begin{bmatrix} 3 & 2 \\ 9 & 5 \end{bmatrix} = (3\times5) - (2\times9)$	-3
M_{23}	$\begin{bmatrix} 3 & 3 \\ 9 & 1 \end{bmatrix} = (3\times1) - (9\times9)$	-78
M_{31}	$\begin{bmatrix} 9 & 2 \\ 4 & 9 \end{bmatrix} = (9\times9) - (2\times4)$	73
M_{32}	$\begin{bmatrix} 3 & 2 \\ 1 & 9 \end{bmatrix} = (3\times9) - (2\times1)$	25
M_{33}	$\begin{bmatrix} 3 & 9 \\ 1 & 4 \end{bmatrix} = (3\times4) - (9\times1)$	3

Bentuk matriks

$M = \begin{bmatrix} 11 & -76 & -35 \\ 43 & -3 & -78 \\ 73 & 25 & 3 \end{bmatrix}$

c. matriks kofaktor, matriks adjoin $Adj(A) = C^T$, $A^{-1} = (1/\det(A)).Adj(A)$ dalam bentuk pecahan

- Matriks kofaktor C

Pola tanda $\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$

$M = \begin{bmatrix} 11 & -76 & -35 \\ 43 & -3 & -78 \\ 73 & 25 & 3 \end{bmatrix}$

$C = \begin{bmatrix} +11 & -(-76) & +(-35) \\ -(43) & +(-3) & -(-78) \\ +73 & -(25) & +3 \end{bmatrix}$

$C = \begin{bmatrix} 11 & 76 & -35 \\ -43 & -3 & 78 \\ 73 & -25 & 3 \end{bmatrix}$

- Matriks Adjoin $Adj(A) = C^T$

Transpose (tukar baris ↔ kolom)

$Adj(A) = \begin{bmatrix} 11 & -43 & 73 \\ 76 & -3 & -25 \\ -35 & 78 & 3 \end{bmatrix}$

- Invers Matriks

$A^{-1} = \frac{1}{\det(A)} \cdot Adj(A)$

$A^{-1} = \frac{1}{647} \begin{bmatrix} 11 & -43 & 73 \\ 76 & -3 & -25 \\ -35 & 78 & 3 \end{bmatrix}$

$A^{-1} = \begin{bmatrix} 11/647 & -43/647 & 73/647 \\ 76/647 & -3/647 & -25/647 \\ -35/647 & 78/647 & 3/647 \end{bmatrix}$

SiDU

e. Blok piksel asli P(3x3)

$$P = \begin{bmatrix} 120 & (50+N) & 200 \\ 80 & a+2 & (10+N) \\ (30+N) & 90 & 170 \end{bmatrix}$$

- Substitusi :
  - $50 + 29 = 79$
  - $a+2 = 4$
  - $10 + N = 39$
  - $50 + N = 59$

$$P = \begin{bmatrix} 120 & 79 & 200 \\ 80 & 4 & 39 \\ 59 & 90 & 170 \end{bmatrix}$$

- Enkripsi $E = A \cdot P$

$$\begin{bmatrix} 3 & 9 & 2 \\ 1 & 4 & 9 \\ 9 & 1 & 5 \end{bmatrix} \cdot \begin{bmatrix} 120 & 79 & 200 \\ 80 & 4 & 39 \\ 59 & 90 & 170 \end{bmatrix}$$

Baris 1

- $(1,1) : 3(120) + 9(80) + 2(59) = 360 + 720 + 118 = 1198$
- $(1,2) : 3(79) + 9(4) + 2(90) = 237 + 36 + 180 = 453$
- $(1,3) : 3(200) + 9(39) + 2(170) = 600 + 351 + 340 = 1291$

Baris 2

- $(2,1) : 1(120) + 4(80) + 9(59) = 120 + 320 + 531 = 971$
- $(2,2) : 1(79) + 4(4) + 9(90) = 79 + 16 + 810 = 905$
- $(2,3) : 1(200) + 4(39) + 9(170) = 200 + 156 + 1530 = 1886$

Baris 3

- $(3,1) : 9(120) + 1(80) + 5(59) = 1080 + 80 + 295 = 1455$
- $(3,2) : 9(79) + 1(4) + 5(90) = 711 + 4 + 450 = 1165$
- $(3,3) : 9(200) + 1(39) + 5(170) = 1800 + 39 + 850 = 2689$

Hasil Enkripsi

$$E = \begin{bmatrix} 1198 & 453 & 1291 \\ 971 & 905 & 1886 \\ 1455 & 1165 & 2689 \end{bmatrix}$$

- Deskripsi : $P' = A^{-1} \cdot E$

$$P' = \frac{1}{647}(Adj(A) \cdot E) = \frac{1}{647} \begin{bmatrix} 11 & -43 & 73 \\ 76 & -3 & -25 \\ -35 & 78 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1198 & 453 & 1291 \\ 971 & 905 & 1886 \\ 1445 & 1165 & 2689 \end{bmatrix}$$

Baris 1

- $(1,1) = \frac{1}{647}(11 \times 1198 - 43 \times 971 + 73 \times 1455)$
  $= \frac{1}{647}(13178 - 41753 + 106215) = \frac{77640}{647} = 120$
- $(1,2) = \frac{1}{647}(11 \times 453 - 43 \times 905 + 73 \times 1165)$
  $= \frac{1}{647}(4983 - 38915 + 85095) = \frac{51113}{647} = 79$
- $(1,3) = \frac{1}{647}(11 \times 1291 - 43 \times 1886 + 73 \times 2689)$
  $= \frac{1}{647}(14201 - 81098 + 196297) = \frac{129400}{647} = 200$

Baris 2

- $(2,1) = \frac{1}{647}(76 \times 1198 - 3 \times 791 - 25 \times 1455)$
  $= \frac{1}{647}(91048 - 2913 - 36375) = \frac{51760}{647} = 80$
- $(2,2) = \frac{1}{647}(76 \times 453 - 3 \times 905 - 25 \times 1165)$
  $= \frac{1}{647}(34428 - 2715 - 29125) = \frac{2588}{647} = 4$
- $(2,3) = \frac{1}{647}(76 \times 1291 - 3 \times 1886 - 25 \times 2689)$
  $= \frac{1}{647}(98116 - 5658 - 67225) = \frac{25233}{647} = 39$

Baris 3

- $(3,1) = \frac{1}{647}(-35 \times 1198 + 78 \times 971 + 3 \times 1455$
  $= \frac{1}{647}(-41930 + 75738 + 4365) = \frac{38173}{647} = 59$
- $(3,2) = \frac{1}{647}(-35 \times 453 + 78 \times 905 + 3 \times 1165)$
  $= \frac{1}{647}(-15855 + 70590 + 3495$
  $= \frac{58230}{647} = 90$
- $(3,3) = \frac{1}{647}(-35 \times 1291 + 78 \times 1886 + 3 \times 2689)$
  $= \frac{1}{647}(-45185 + 147108 + 8067$
  $= \frac{109990}{647} = 170$

Hasil Akhir

$$P' = \begin{bmatrix} 120 & 79 & 200 \\ 80 & 4 & 39 \\ 59 & 90 & 170 \end{bmatrix}$$

Terbukti $P' = P$

SiDU

2. e. Diskusikan apa yang terjadi jika det(A) = 0 ?

Jika det(A) = 0, maka matriks A tidak memiliki invers sehingga proses deskripsi $P' = A^{-1} E$ tidak dapat dilakukan. Selain itu, transformasi menjadi tidak satu-ke-satu (many-to-one). menyebabkan informasi hilang dan data asli tidak dapat dipulihkan. Oleh karena itu, matriks dengan determinan nol tidak dapat digunakan sebagai kunci enkripsi.